Machine Learning in Time Series

Lecture 2: Kernel Methods, SVM, and Support Vector Regression (SVR)

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Spring 2026

PhD Seminar in Econometrics / Data Science
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Agenda

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Seminar Overview & Roadmap

The 6-Lecture Journey

Where we are

In Lecture 1, we tackled regression in high-dimensional spaces ($p \gg T$) using penalized objective functions (Ridge/Lasso) to control model variance. Today, we shift to **Classification**, **Non-linear Mapping**, and **Support Vector Regression**.

Lecture 1: Historical Evolution, the Prediction Problem, and Shrinkage.

Lecture 2: Kernel Methods, SVM, and SVR.

Lecture 3: High-Dimensional Linear Dynamics (BVAR, Sparse VAR).

Lecture 4: Deep Learning for Sequences (RNN/LSTM/Attention).

Lecture 5: Trees, Boosting, and Ensembles.

Lecture 6: NLP, Attention, and Causal ML.

The Classification Problem in Macro/Finance

Why Do Economists Need Classification?

Not all economic questions involve predicting a continuous mean. Many critical macroeconomic and financial phenomena are fundamentally **discrete states or regimes**.

The Need to Predict Binary Outcomes ($y \in \{0, 1\}$ or $\{-1, 1\}$):

- **Business Cycles:** Is the economy in an expansion (0) or a recession (1)?
- **Financial Markets:** Will the stock market go up ($+1$) or down (-1) next month?
- **Crisis Events:** Will a sovereign nation default on its debt next year?
- **Monetary Policy:** Will the Central Bank hike interest rates at the next meeting?

We can use quantitative data (yield curves, inflation, unemployment) as features X, but our target y is a categorical regime.

The Classical Approach: Logistic Regression

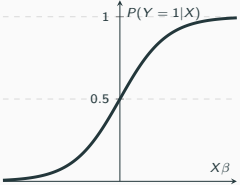
The Genesis of Logit

The Problem with OLS for Binary Data: If we use standard OLS regression ($y = \beta_0 + \beta_1 X$) on a binary target (0 or 1), the predicted line will eventually shoot past 1 or drop below 0. Predicting a probability of 1.5 or -0.2 is mathematically nonsensical.

The Logit Solution: To ensure predictions stay bounded between 0 and 1, we map the linear combination of inputs through the **Logistic (Sigmoid) Function**:

$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

The Sigmoid S-Curve



The S-curve elegantly compresses any input value from $-\infty$ to $+\infty$ into a valid probability space $[0, 1]$.

How Logit Works Mathematically

With algebraic manipulation, we can rewrite the Sigmoid function to reveal its core econometric intuition: **The Log-Odds Ratio**.

$$\log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k$$

Interpreting Logit in Economics

- The left side is the natural log of the odds of an event occurring (e.g., the odds of a recession).
- The right side is a familiar linear equation.
- **Interpretation:** A 1-unit increase in X_1 increases the *log-odds* of the event by β_1.

The Critical Limitation: While the probabilities are S-shaped, the decision boundary dividing the classes is still strictly **linear**. It struggles to capture complex, non-linear interactions without massive manual feature engineering.

1. The ROC Curve: Visualizing Thresholds

Logistic regression outputs a **probability** (0 to 1). We must choose a **threshold** (e.g., 0.5) to classify the result as Yes or No.

The **ROC Curve** graphs the model's performance across *all possible thresholds* using two metrics:

- **Y-Axis: True Positive Rate (TPR)**
  Also called Sensitivity. The percentage of actual positive cases we correctly identified. (We want this high!)
- **X-Axis: False Positive Rate (FPR)**
  Also called 1 - Specificity. The percentage of actual negative cases we accidentally flagged as positive. (We want this low!)

2. Interpreting the Curve & AUC

How do we know if our model is actually good?

Reading the Graph:

- **Top-Left Corner:** The ideal spot (100% TPR, 0% FPR). We want our curve to "hug" this corner as tightly as possible.
- **Diagonal Line:** Represents a completely useless model that is just guessing randomly.

AUC (Area Under the Curve)

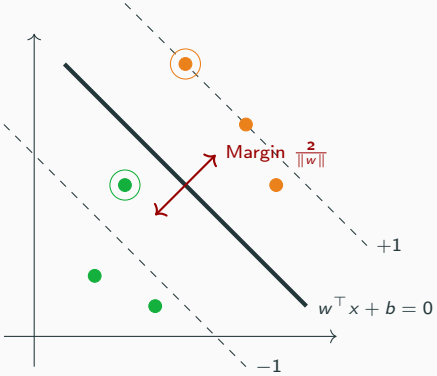
A single number summarizing the entire ROC curve:

- **AUC = 1.0:** A perfect model.
- **AUC = 0.8 to 0.9:** A very good model.
- **AUC = 0.5:** A useless model (random guessing).

Support Vector Machines: The Primal Problem

The Geometry of Maximum Margin

Support Vector Machines (SVM) take a geometric approach. They seek the hyperplane that perfectly separates the classes with the **widest possible margin**.



Key Definitions:

- **Hyperplane:** $w^\top x + b = 0$.
- **Support Vectors:** The data points sitting exactly on the margin boundaries ($y_i(w^\top x_i + b) = 1$).
- **Margin Width:** Maximizing the margin is geometrically equivalent to minimizing $\frac{1}{2}\|w\|^2$.

Notice that the decision boundary depends ONLY on the support vectors. All other data points can be removed without changing the model!

The Soft Margin and the C Parameter

We formulate the search for the optimal hyperplane as a constrained quadratic optimization problem. To handle real-world noise, we introduce **slack variables** ($\xi_i \geq 0$) that allow some points to violate the margin, subject to a penalty C.

The Soft Margin Primal Formulation

$$\min_{w,b,\xi} \left(\frac{1}{2}\|w\|^2 + C \sum_{i=1}^{N} \xi_i \right)$$

$$\text{subject to:} \quad y_i(w^\top x_i + b) \geq 1 - \xi_i \quad \text{and} \quad \xi_i \geq 0$$

The C Hyperparameter (Bias-Variance Tradeoff):

- **High C:** Heavy penalty for errors. Hard margin. High variance (overfitting).
- **Low C:** Soft penalty. Wider margin, more violations allowed. High bias, robust to noise.

Lagrange and the Dual Formulation

The Lagrangian Formulation

We introduce Lagrange multipliers $\alpha_i \geq 0$ to incorporate the constraints into the objective function.

The Lagrangian Function

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2}\|w\|^2 - \sum_{i=1}^{N} \alpha_i \left[y_i(w^\top x_i + b) - 1\right]$$

By taking the partial derivatives with respect to w and b and setting them to zero, we obtain the KKT conditions:

$$\frac{\partial \mathcal{L}}{\partial w} = 0 \implies w = \sum_{i=1}^{N} \alpha_i y_i x_i$$

$$\frac{\partial \mathcal{L}}{\partial b} = 0 \implies \sum_{i=1}^{N} \alpha_i y_i = 0$$

The Dual Formulation

Substituting the KKT conditions back into the Lagrangian yields the **Dual Formulation**.

The Dual Problem

$$\max_{\alpha} \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j y_i y_j \underbrace{(\mathrm{x_i}^\top \mathrm{x_j})}_{\text{Dot Product}}$$

$$\text{subject to: } \sum_{i=1}^{N} \alpha_i y_i = 0 \quad \text{and} \quad 0 \leq \alpha_i \leq C$$

Why is this a breakthrough? The data features (x) now *only* appear inside a dot product ($x_i^\top x_j$), which is simply a measure of **similarity** between two observations.

Non-Linearity: Kernels and Transformations

Transforming Variables: The OLS Parallel

In traditional Econometrics, if a relationship isn't linear, what do we do? We manually transform the variables.

Example: Predicting Output (y) based on Labor (L) and Capital (K). If the relationship is a Cobb-Douglas production function ($y = A \cdot L^{\alpha} K^{\beta}$), running a simple linear OLS ($y = \beta_1 L + \beta_2 K$) will fail miserably.

The Econometric Fix (Manual Mapping): We create new, transformed variables by taking the natural log:

$$\ln(y) = \ln(A) + \alpha \ln(L) + \beta \ln(K)$$

By mapping our original features (L, K) into a new feature space ($\ln L, \ln K$), a non-linear problem becomes perfectly linear!

What is a Kernel? A Layman's Explanation

The Flat Sheet of Paper Analogy

Imagine you have a flat sheet of paper. You draw red dots in the center and blue dots around the edges. If I ask you to separate the red and blue dots using only a single, perfectly straight cut with scissors, you can't do it. The problem is **not linearly separable** in 2D.

Now, imagine picking the paper up and pushing the center up from underneath, warping it into a 3D cone shape (the red dots are now at the peak, the blue dots are at the base).

Now, you can take a flat knife, slice horizontally through the 3D cone, and easily separate the red peak from the blue base.

The "Trick": A **Kernel** is mathematical instructions for how to warp that paper. It calculates the 3D distance between points without us having to physically fold the paper or compute all those extra dimensions.

Formalizing Kernel Functions

A Kernel is a function that takes two vectors u and v and returns their inner product in some high-dimensional feature space: $K(u, v) = \phi(u)^\top \phi(v)$. Using Kernels, we can efficiently perform classification in high-dimensional spaces without explicitly transforming the data.

Standard Kernel Functions:

- **Linear:** $K(u, v) = u^\top v$
- **Polynomial:** $K(u, v) = (\gamma u^\top v + r)^d$
- **Radial Basis Function (RBF):** $K(u, v) = e^{-\gamma \|u-v\|^2}$
- **Sigmoid:** $K(u, v) = \tanh(\gamma u^\top v + r)$

Transitioning to Support Vector Regression (SVR)

From Classification to Regression (SVM to SVR)

The Transition

In SVM, we sought a hyperplane that separated two classes with a maximum margin. In **Support Vector Regression (SVR)**, we flip the logic: we seek a hyperplane (a regression line) that fits the continuous data *inside* a defined margin, while remaining as flat as possible.

Instead of minimizing the sum of squared errors (like OLS), SVR introduces the ϵ-**insensitive loss function**.

- We define a margin of tolerance, ϵ.
- If a prediction error is smaller than ϵ, the loss is exactly **zero**. The model ignores tiny, noisy fluctuations.
- Errors outside the ϵ-tube are penalized linearly, making SVR highly robust to macroeconomic outliers.

1. The ϵ-Tube: The "Close Enough" Zone

Quote: *"On the phrase "Errors outside the ϵ-tube are penalized linearly, making SVR highly robust to macroeconomic outliers."*

How standard regression and SVR view errors differently:

- **Standard Regression:** Tries to hit every point perfectly. Any deviation, even a millimeter, is counted as an error.
- **SVR (ϵ-Tube):** Draws a prediction line, but adds a buffer zone (tube) of width ϵ around it.
- **Epsilon-Insensitive Loss:** If a data point falls *anywhere inside* this tube, the model considers the error to be **zero**.

The Takeaway

SVR doesn't sweat the small stuff. If predictions are within the ϵ boundary, the model is satisfied.

2. Linear Penalties & Macroeconomic Outliers

Quote: *"On the phrase "Errors outside the ϵ-tube are penalized linearly, making SVR highly robust to macroeconomic outliers."*

What happens when a point falls *outside* the tube?

- **Standard Regression (OLS):** Errors are **squared**. An outlier 10 units away creates a massive penalty of 100, forcing the line to bend drastically toward the anomaly.
- **SVR:** Errors are penalized **linearly**. An outlier 10 units away outside the tube gets a penalty of exactly 10.

Why it's "Robust"

Macroeconomic data (like market crashes or supply shocks) has wild outliers. Because SVR doesn't square the errors, it doesn't panic. The main prediction line remains stable despite extreme anomalies.

SVR: The Primal Optimization Problem

To allow for errors outside the ϵ-tube, we introduce two slack variables: ξ_i (for errors above the tube) and ξ_i^* (for errors below the tube).

SVR Primal Formulation

$$\min_{w,b,\xi,\xi^*} \left(\frac{1}{2}\|w\|^2 + C \sum_{i=1}^{N} (\xi_i + \xi_i^*) \right)$$

$$\text{subject to:} \quad \begin{cases} y_i - (w^\top x_i + b) \leq \epsilon + \xi_i \\ (w^\top x_i + b) - y_i \leq \epsilon + \xi_i^* \\ \xi_i, \xi_i^* \geq 0 \end{cases}$$

Here, C determines the trade-off between the flatness of the model ($\|w\|^2$) and the amount up to which deviations larger than ϵ are tolerated.

SVR: The Dual Formulation

Just like in classification, we use Lagrange multipliers (α_i, α_i^*) to solve the primal problem, leading to the Dual SVR formulation.

SVR Dual Formulation

$$\max_{\alpha,\alpha^*} \left(-\frac{1}{2} \sum_{i,j=1}^{N} (\alpha_i - \alpha_i^*)(\alpha_j - \alpha_j^*) K(x_i, x_j) - \epsilon \sum_{i=1}^{N} (\alpha_i + \alpha_i^*) + \sum_{i=1}^{N} y_i (\alpha_i - \alpha_i^*) \right)$$

$$\text{subject to: } \sum_{i=1}^{N} (\alpha_i - \alpha_i^*) = 0 \quad \text{and} \quad \alpha_i, \alpha_i^* \in [0, C]$$

The final regression function takes the form: $f(x) = \sum_{i=1}^{N} (\alpha_i - \alpha_i^*) K(x_i, x) + b$. *Note the Kernel Trick is fully applicable here!*

Lecture 2 Mini-Labs: Classification & Regression

Mini-Lab 1: Market Direction Classification

The Task: Predicting Stock Market Direction

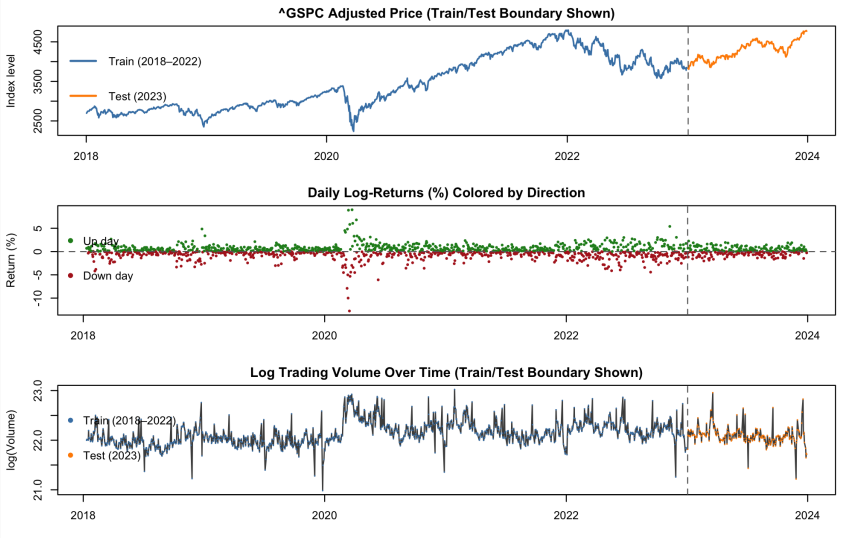
Predict if the S&P 500 will have a positive or negative return today, based on yesterday's returns. **Target:** Direction (Up/Down).

The Traditional Approach (Logit): Historically, we estimate this using Logistic Regression.

- Logit models the probability of the market going 'Up' using 'Lag1', 'Lag2', etc., as linear predictors.
- However, financial returns are notoriously non-linear and noisy. A purely linear decision boundary often fails.

The Lab: Compare OOS accuracy and ROC curves of Logit vs. an SVM with an RBF Kernel.

Mini-Lab 1: Market Direction Classification



Parameter Tuning & K-Fold Cross Validation

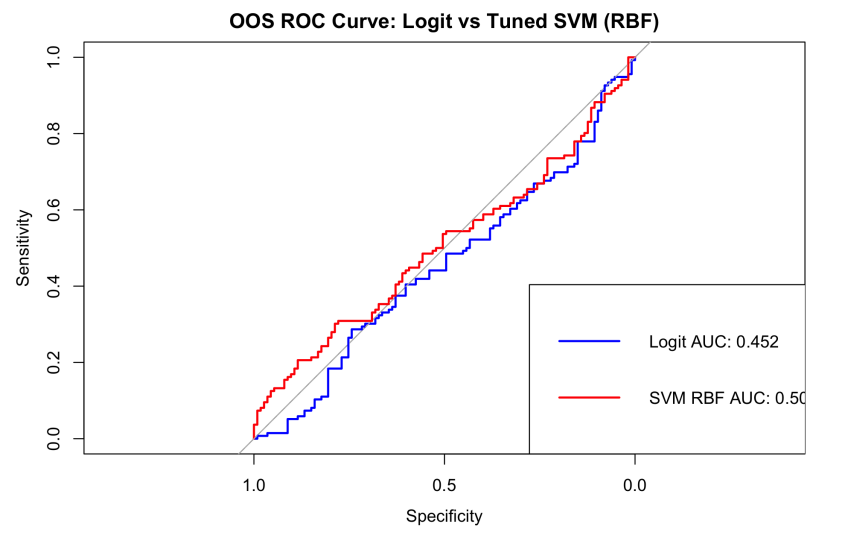
For SVR (and SVM), we must tune hyperparameters: C (cost), ϵ (tube width), and γ (RBF kernel parameter). We do this using **K-Fold Cross-Validation**.

How K-Fold CV Works:

1. Divide the training data into K equal-sized subsets (folds).
2. Train the SVR model on $K-1$ folds for a specific combination of (C, ϵ, γ).
3. Validate the model on the single hold-out fold and record the error (e.g., RMSE).
4. Repeat K times, and average the errors. Select the parameters with the lowest average error.

Time Series Caveat: Standard K-fold randomizes data, breaking temporal dependence. For macro forecasting, we must use **time-series blocked CV** or rolling/expanding windows to prevent data leakage!

Mini-Lab 1: Market Direction Classification



Reading Discussion: Gu, Kelly, & Xiu (2020)

The Big Question: Can ML predict a Random Walk?

If a financial time series is a *true* Random Walk ($y_{t+1} = y_t + \epsilon_{t+1}$), no algorithm on Earth can predict it. By definition, the next step is pure noise. ML is not magic; it cannot predict the unpredictable.

So why did Gu, Kelly, and Xiu find that ML successfully predicts stock returns?

- **The Reality of Markets:** Financial returns are not pure random walks; they contain structural risk premia and behavioral inefficiencies. However, they have an exceptionally **low signal-to-noise ratio**.
- **The Failure of OLS:** When dealing with 99% noise and 1% signal across hundreds of predictors, classical linear models inevitably overfit the noise and collapse out-of-sample.
- **The ML Solution:** ML models succeed because **regularization and non-linear mapping** allow them to aggregate very faint, complex signals across massive datasets while mathematically refusing to memorize the noise.

Mini-Lab 2: Forecasting Inflation with SVR

The Task: The Project Continued

Forecast continuous US YoY Core CPI Inflation using an AR(12) feature matrix. We will compare OLS against tuned Support Vector Regression (SVR).

Pipeline Requirements:

1. **Data Prep:** Construct the lagged feature matrix. Apply a strict chronological train/test split.
2. **Tuning SVR:** Use time-series CV to tune C, γ, and ϵ on the training data.
3. **Evaluate:** Fit an SVR with `kernel="radial"`. Compare its OOS RMSE to classical OLS. Does the non-linear kernel and ϵ-tube improve robustness against inflation shocks?

Lab Results: SVR vs. OLS for Inflation

Model	OOS RMSE
1. Standard AR(12) (OLS)	0.1866
2. Linear SVR (ϵ-tube)	0.1928
3. RBF Kernel SVR (Non-linear)	0.2601

Pedagogical Takeaways: Why did OLS win?

- **The Linearity of Inflation:** The short-term AR(12) relationship for YoY CPI is predominantly linear.
- **The Cost of Unnecessary Complexity:** The RBF Kernel maps data into infinite dimensions to find complex, non-linear patterns. When the true Data Generating Process is linear, RBF simply overfits the training noise, causing OOS RMSE to degrade.
- **SVR vs OLS:** The ϵ-tube in Linear SVR adds robustness against outliers. However, if the test period lacks massive structural shocks, OLS remains the most statistically efficient linear estimator.

Advanced Lab: Rolling Windows & Macro Variables

Methodology: Rolling Window Cross-Validation

The Problem with Fixed Splits in Time Series

A static 80/20 train-test split might land the test set entirely inside a specific economic regime (e.g., the post-COVID inflation spike), making it impossible to tell if the model is robust or just lucky.

The Solution: Rolling Windows

- **Training Window:** We use a fixed rolling window of 120 months (10 years) to train the models.
- **Forecasting Step:** We predict only the single next month ($t + 1$).
- **Slide and Repeat:** The window shifts forward one month, and we retrain all models from scratch. We repeat this for the last 60 months (5 years) of data.

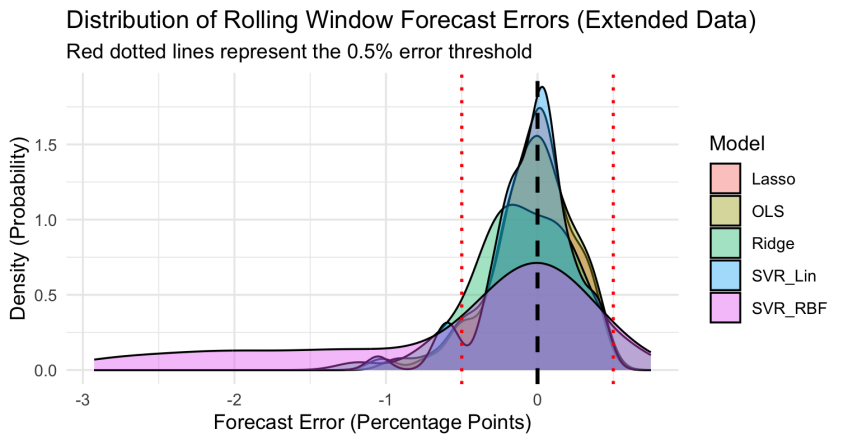
This perfectly simulates how a quantitative researcher makes real-time forecasts month over month!

Results: Out-Of-Sample Rolling Metrics

Target: Predict YoY Core CPI.
Metric 1: RMSE **Metric 2:** Prob($|\text{Error}| > 0.5\%$)

	Base (AR-12 Only)		Extended (+ Macro Vars)	
Model	**RMSE**	**Prob($> 0.5\%$)**	**RMSE**	**Prob($> 0.5\%$)**
OLS	0.2850	8.3%	0.2777	8.3%
Ridge	0.3624	11.7%	0.3597	13.3%
Lasso	0.2841	6.7%	0.2800	**6.7%**
SVR (Linear)	**0.2692**	**5.0%**	0.2794	8.3%
SVR (RBF)	1.0085	30.0%	1.0608	30.0%

Visualizing the Error Distribution



Note: SVR (Linear) and Lasso have the tightest distributions centered around zero, remaining well inside the 0.5% error threshold (red dotted lines).

Pedagogical Takeaways: What did we learn?

1. **Linear SVR shines on the Base Data:** The linear ϵ-tube model achieved the absolute best error probability (5.0%) on the AR-12 data. It successfully ignored tiny inflation noise while capturing the momentum.
2. **More variables $\neq$ Better Forecasts:** Adding Fed Funds, S&P 500, and Industrial Production actually *hurt* the Linear SVR and Ridge models. The macro variables introduced more noise than signal.
3. **Lasso's Feature Selection:** Lasso tied for the best extended-data model (6.7% error prob). Why? Because it likely forced the coefficients of the noisy macroeconomic variables to exactly zero, essentially reverting to the cleaner AR structure.
4. **The RBF Disaster:** The non-linear SVR completely failed (30% large error probability). It vastly overfit the training data's noise, proving that complex non-linear kernels are dangerous when the true DGP is mostly linear.

Concluding Remarks

Synthesis: Key Takeaways from Lecture 2

1. **The Margin beats the Likelihood:** SVMs focus only on the hardest-to-classify points (Support Vectors) by maximizing the geometric margin, making them robust in complex spaces.
2. **Non-linearity for free:** The Dual formulation allows us to replace dot products with Kernel functions (like RBF), mapping data into infinite-dimensional spaces without the computational curse.
3. **Robust Regression:** SVR replaces squared loss with an ϵ-insensitive tube, ignoring minor noise and penalizing outliers linearly rather than quadratically.
4. **Tuning is paramount:** An SVM/SVR with poorly tuned parameters is a terrible model. Time-aware cross-validation is strictly required to find the optimal C, γ, and ϵ.

Looking Ahead: Lecture 3

Next Session: High-Dimensional Linear Dynamics

In Lecture 3, we return to forecasting continuous macroeconomic targets, scaling up the "system" view using Vector Autoregressions (VAR) under $p \gg T$ constraints.

Topics to prepare for:

- Bayesian VARs (BVAR) and Minnesota priors interpreted as a form of shrinkage.
- Sparse VAR penalties (Lasso / Group Lasso applied to multivariate systems).
- Forecasting vs. interpretability and uncertainty quantification.

Reminder: Reading memos for Giannone et al. (2015) and Nicholson et al. (2020) are due before the next class!